Cardiovascular System Modeling using 0-Dimensional Models

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Background

History of Lumped Parameter Modeling

0-Dimensional/lumped parameter models provide a concise way to describe blood flow interactions in the cardiovascular system

1899: Otto Frank creates the 2-element Windkessel Model

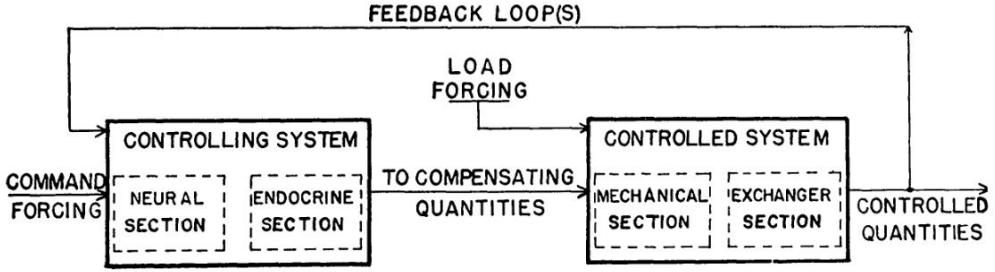
1940s - 1950s: Many models are created based off of the Windkessel Model

1959: Arthur Grodins creates a closed circuit model of the cardiovascular system based on control theory

Purpose: Explore the Grodins model to understand cardiac hemodynamics through pressure regulation modeling

The Grodins Model

- Simulate steady-state conditions of the cardiovascular system
- Attempt to model the cardiovascular system as a feedback regulator
- Quantities of the controlled system are fed back to the controlling system whose outputs influence properties of the controlled system



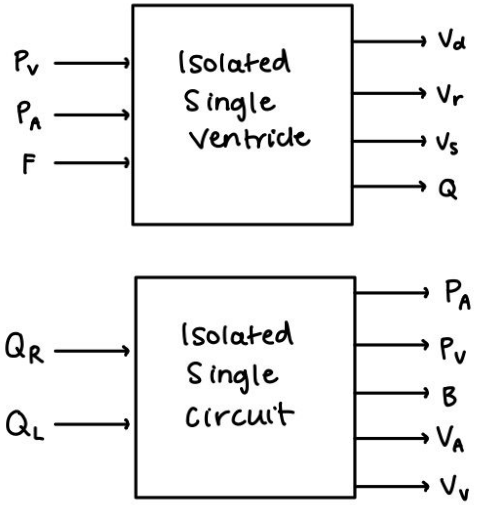
The Grodins Model

Isolated Compartments

- The cardiovascular regulator is split into four components:
  - Left and right ventricular pumps
  - Pulmonary circuit
  - Systemic circuit

Method:

1. Isolate each component
2. Identify inputs and outputs of each
3. Determine the relationship between the inputs and outputs to obtain governing equations

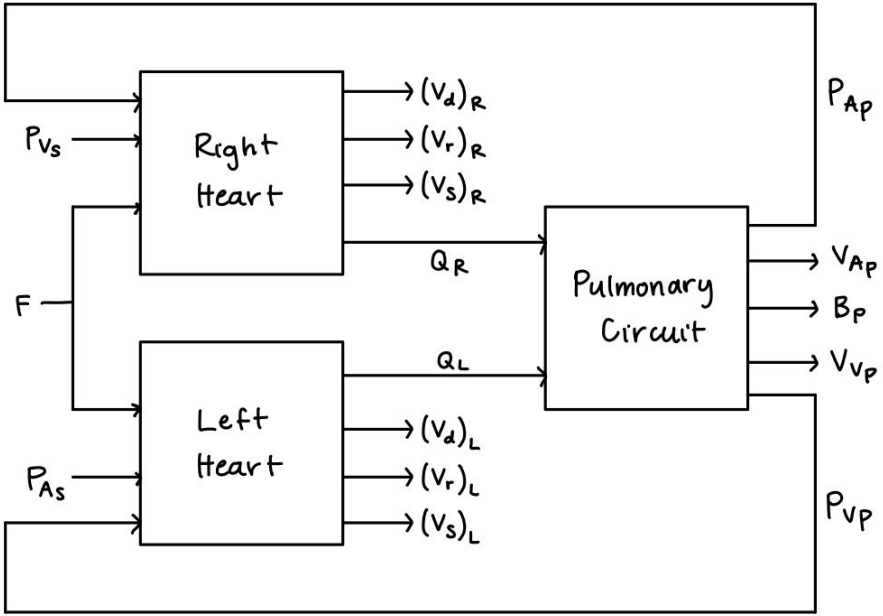


- P_v, P_A = venous & arterial pressures
- V_d, V_r, V_s = ventricular volumes (diastolic, stroke, residual)
- F = cardiac frequency
- V_A, V_V = arterial and venous blood volumes
- B = total blood volume

The Grodins Model

The Starling Heart-Lung Preparation

- Flow out of the right heart must match flow into the left heart or flow will accumulate or be drawn out of the lungs

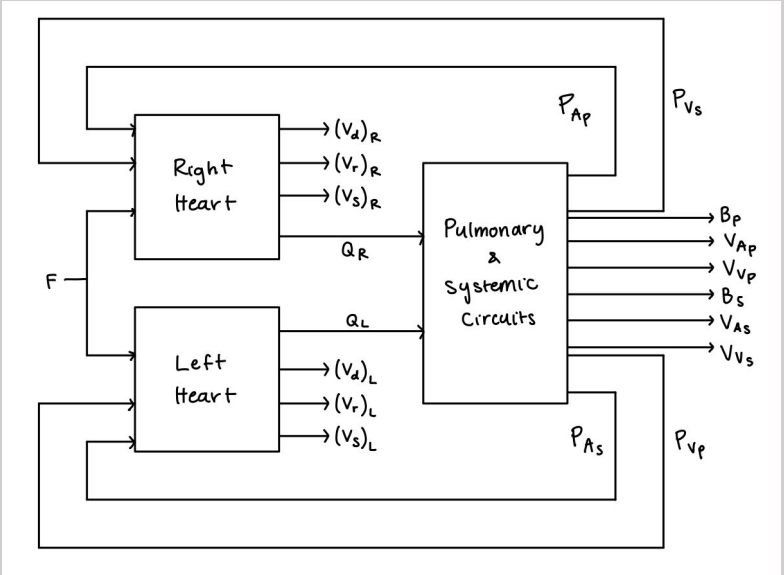


- $P_{V,s}$, $P_{A,s}$ = venous filling pressure & arterial load pressure (systemic)
  - Independent variables
- $P_{V,p}$, $P_{A,p}$ = venous filling pressure & arterial load pressure (pulmonary)
  - Dependent variables

The Grodins Model

The Full Model

- The final model incorporates the two pumps, the pulmonary circuit and the systemic circuit
- Must define cardiac and circuit properties to obtain remaining independent variables
- Cardiac properties:
  - S_L, S_R = left and right ventricle 'strength'
  - C_L, C_R = left and right compliance of the relaxed ventricle
  - R_L, R_R = left and right resistance to filling
- Circuit properties:
  - R_p, R_s = pulmonary and systemic arteriole resistance
  - $C_{A,s}$, $C_{V,s}$, $C_{A,p}$, $C_{V,p}$ = arterial and venous systemic and pulmonary compliances
  - B = total blood volume



The Grodins Model

Obtaining Cardiac and Circuit Properties

The Ventricle

- Steady-state operation with emptying and filling processes

Emptying Process

- Isolate a single ventricular systole
- Define 'stroke work': $w_s = SV_d$
- Then the emptying process is defined by:
  - $V_s = (S/P_A)V_d$
  - $V_r = [1 - (S/P_A)]V_d$

Filling Process

- Isolate a single ventricular diastole
- Filling law:
  - $R\dot{C}V_d + V_d = CP_V$
  - $V_d = CP_V + (V_r - CP_V)\exp(-t/RC)$
  - Initial condition $V_d = V_r$
- $Q = F^*S^*C^*A^*P_V/AP_A + SK$
  - A, K = cardiac frequency parameters

The Circuit

- Systemic circuit equations:
  - $F_S = (P_{A,s} - P_{V,s})/R_S$ = systemic flow rate of blood
  - $P_{A,s} = (1/C_{A,s})V_{A,s}$
  - $P_{V,s} = (1/C_{V,s})V_{V,s}$
  - $B_S = V_{A,s} + V_{V,s}$
- Pulmonary circuit equations:
  - $F_p = (P_{A,p} - P_{V,p})/R_p$ = pulmonary flow rate of blood
  - $P_{A,p} = (1/C_{A,p})V_{A,p}$
  - $P_{V,p} = (1/C_{V,p})V_{V,p}$
  - $B_p = V_{A,p} + V_{V,p}$
- Steady state restrictions:
  - $Q_L = Q_R$, $Q_L = F_s$, $Q_R = F_p$

The Full System = 23 equations, 23 unknowns, 14 independent variables

The Grodins Model – Myocardial Weakness

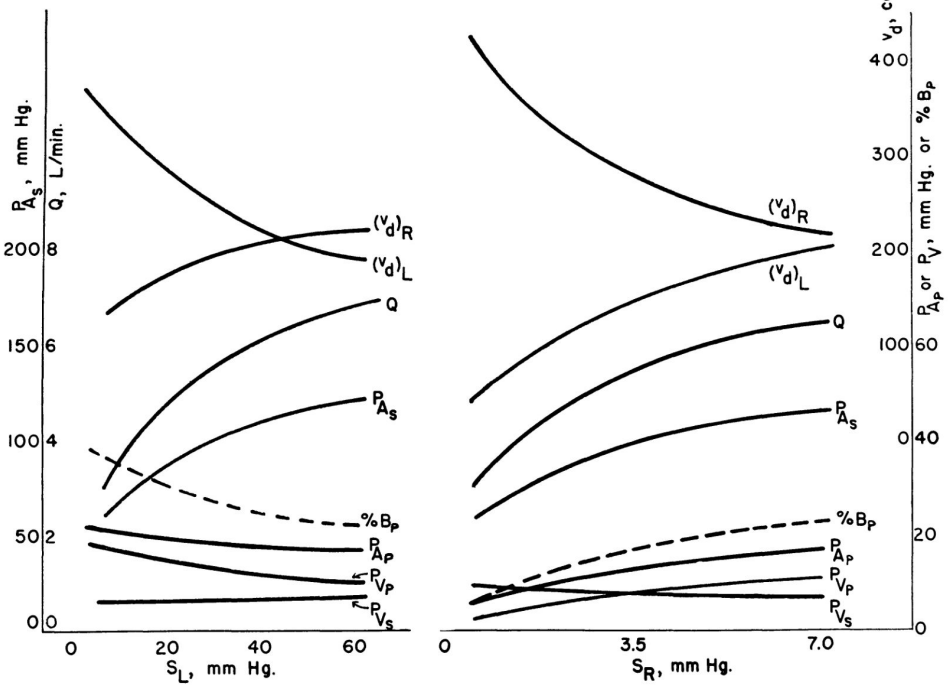


FIG. 20. THEORETICAL RESPONSES, LEFT, TO REDUCTION IN S_L, AND RIGHT, TO REDUCTION IN S_R

- In myocardial weakness the strength of either ventricle is reduced
- The Grodins model for myocardial weakness was in great agreement with observations made by Starr and Rawson using a mechanical model

Discussion

Comparison with the Windkessel Model

Basic Circuitry

- Lumped Parameter Modeling allowing for cheap calculations with ODE's

Single vs Multi-Compartment

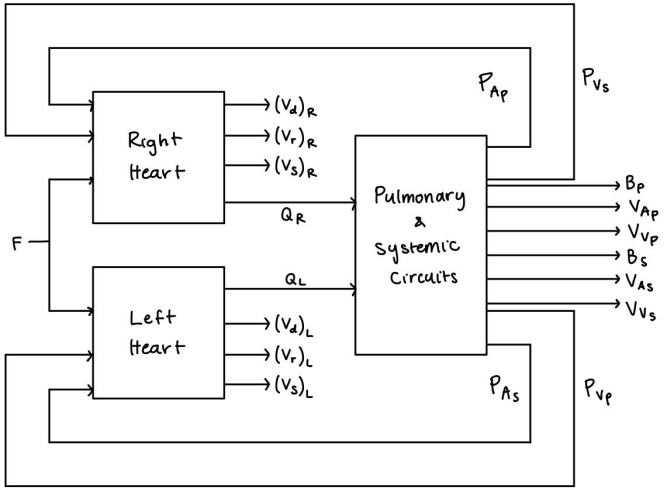
- Greater accuracy and ability for specified R,C values

Baroreflex Mechanisms

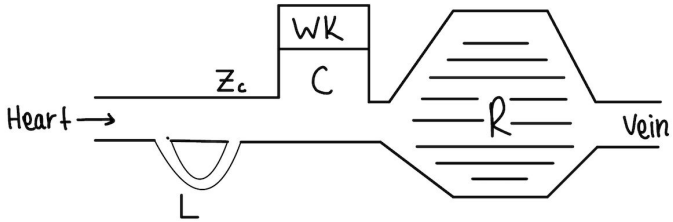
- Grodins's model is capable of showing the pressure regulation of the baroreceptors on the arterial blood pressure

Interrelationship modeling

- Arterial pressure, heart rate, systemic resistance, and cardiac output for both cardiac chambers



Grodins's Model



4 - Element Windkessel Model

Concluding Remarks

Concepts underlying the Grodin model can be used in developing new complex cardiovascular models

- Kappel and Peer, 1993
  - Adapted Grodins's model to study physiological transitions under a workload
- Batzel et al., 2007
  - Utilizes a single feedback loop with a baroreflex control that stabilizes blood pressure and heart rate

References

Curcio, Luciano, et al. “Seven Mathematical Models of Hemorrhagic Shock.” *Computational and Mathematical Methods in Medicine*, vol. 2021, 2021, pp. 1–34., https://doi.org/10.1155/2021/6640638.

Grodins, Fred S. “Integrative Cardiovascular Physiology: A Mathematical Synthesis of Cardiac and Blood Vessel Hemodynamics.” *The Quarterly Review of Biology*, vol. 34, no. 2, 1959, pp. 93–116., https://doi.org/10.1086/402631.

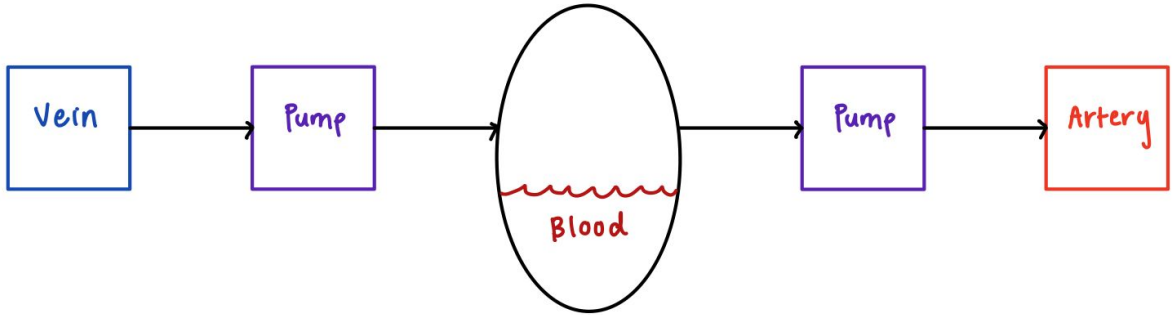
Westerhof, Nico et al. “The arterial Windkessel.” *Medical & biological engineering & computing* vol. 47,2 (2009): 131-41. https://doi:10.1007/s11517-008-0359-2

Thank you!

Backup Slides

Open Volume Concept

- One pump gives blood to the artery from an external infinite blood source
- Another pump takes blood from the central vein and drains it to an external source
- Allows for a transient inequality between the pump outputs
- Total blood volume becomes dependent on other cardiac quantities



The Grodins Model – Changes in Blood Volume

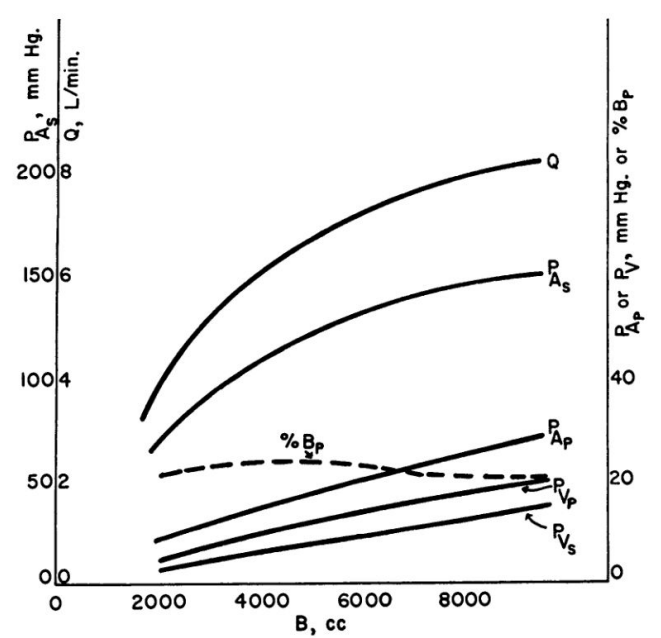


FIG. 14. THEORETICAL RESPONSES TO CHANGES IN BLOOD VOLUME

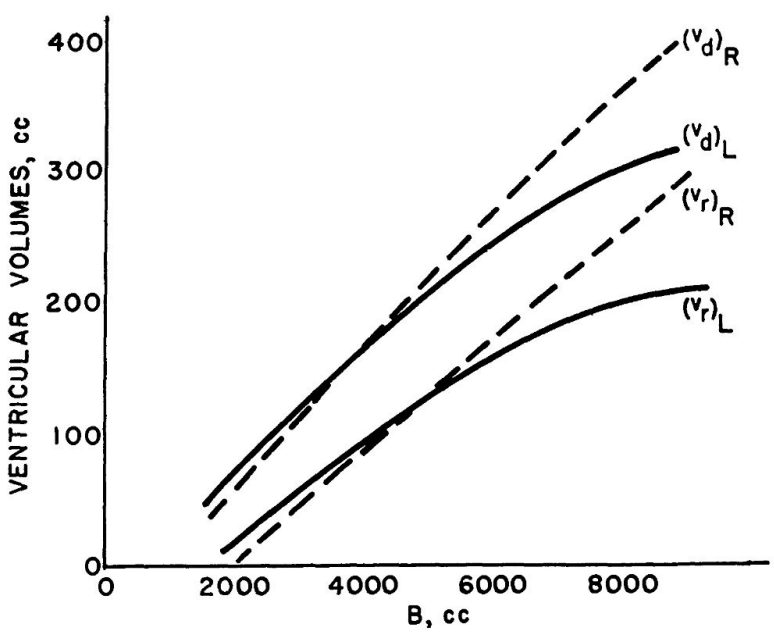
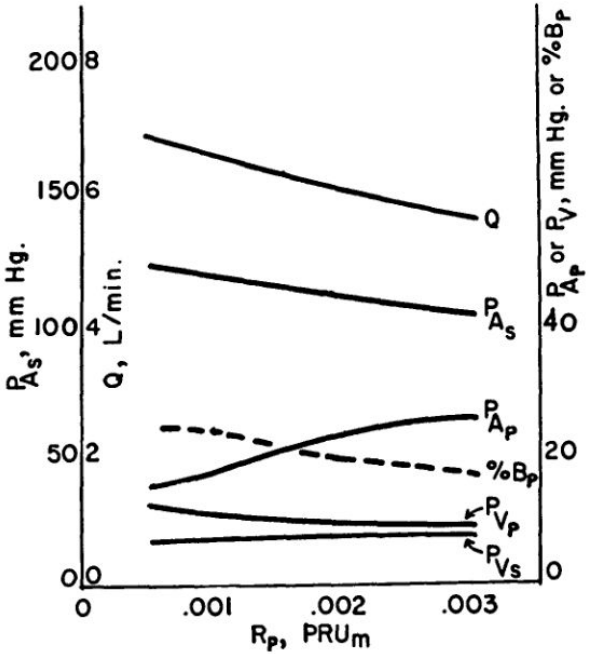


FIG. 15. THEORETICAL RESPONSE OF VENTRICULAR VOLUMES TO CHANGES IN BLOOD VOLUME

The Grodins Model – Changes Peripheral Resistance



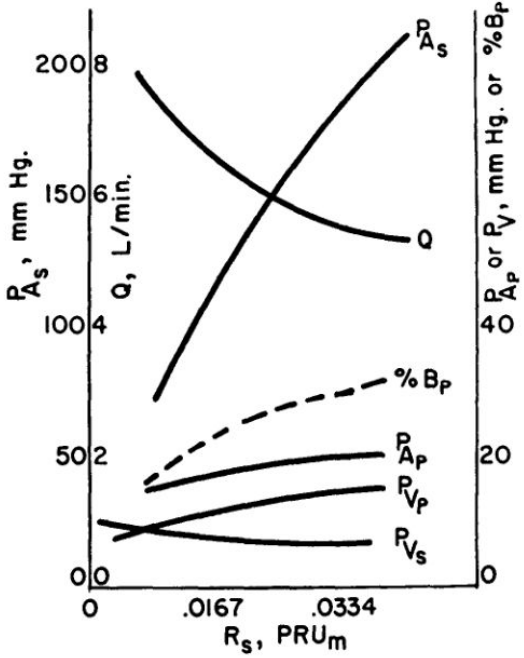


FIG. 16. THEORETICAL RESPONSES, LEFT, TO CHANGES IN R_P, AND RIGHT, TO CHANGES IN R_S

The Grodins Model – Changes Peripheral Resistance

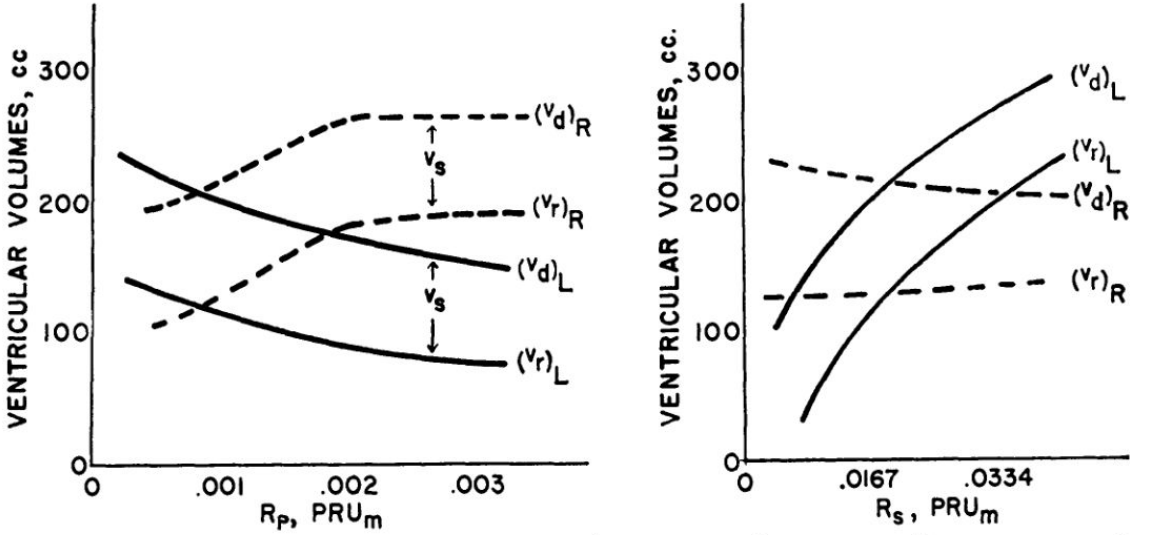


FIG. 17. THEORETICAL RESPONSE OF VENTRICULAR VOLUMES, LEFT, TO CHANGES IN R_P, AND RIGHT, TO CHANGES IN R_S

Grodins Model Strengths and Limitations

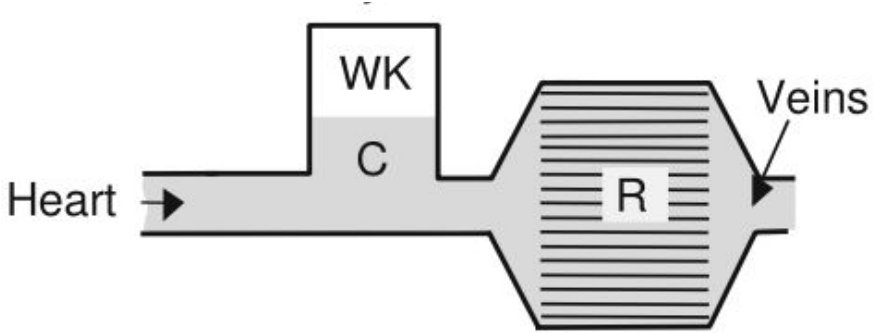
Strengths

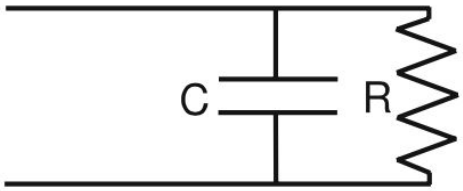
- Most Comprehensive Model
- Relatively Cheap

Limitations

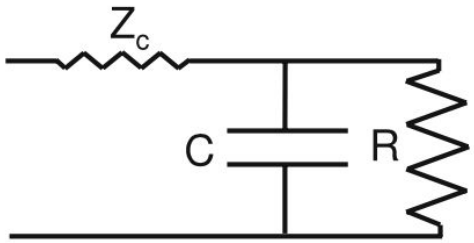
- Does not include Pulsatile flow

Background: The Windkessel Model

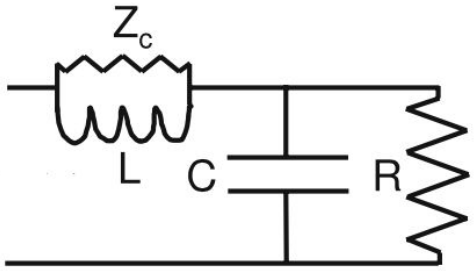




2 - Element



3 - Element



4 - Element 4